ScholAR: Page-Grounded Retrieval-Augmented Generation for Scientific Document Comprehension

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Abstract

Understanding a scientific paper means being able to trace each claim back to the exact place it came from. General-purpose chat over an uploaded PDF is fluent but unreliable for this: nothing ties a cited page number to the text the model was actually shown, and the scientific retrieval-augmented systems that do better tend to rely on cloud-hosted frontier models or datastores of millions of papers. We present ScholAR (**S**mart **C**ompanion for **H**olistic **O**rganization, **L**iterature **A**nalysis & **R**esearch), a retrieval-augmented generation system for scientific documents that runs on a consumer laptop through a local model server, with no cloud API for text or vision (only public paper lookups touch the network). Two design choices carry the system. Page-preserving chunking keeps every retrieved unit inside a single PDF page, so a citation resolves to one highlightable location. Indirect citation grounding lets the model cite only evidence identifiers that map to chunks it actually retrieved, so it cannot write a page number of its own. We evaluate retrieval, answer faithfulness, visual grounding, and cross-document localization on labeled benchmarks. Scored by a local entailment judge, which a negative control shows detects the contradictions a similarity proxy misses, generated answers reach a mean faithfulness of 0.61 across four local models on the diverse benchmark; about a quarter contain a contradicted claim and roughly 9% of citations are unsupported, numbers a cosine proxy had put at 0.85 with none unsupported. Cited pages are valid by construction but support their claim only 66% of the time, no better than a free-form baseline (71%): the mechanism buys provenance, not faithfulness. BM25-primary retrieval reaches Recall@5 of 0.93 across 25 papers (within noise of plain BM25 and a keyword baseline), and figure and table questions route correctly in all 18 pilot cases. On M3SciQA's cross-document localization task, retrieving from the question alone sits near chance, but resolving the anchor figure with the same local multimodal model first reaches an MRR of 0.474, close to GPT-4o (0.500) and ahead of the open-source baselines reported there (on a different split), while running on one machine. Against local reimplementations of PDF-chat, vanilla RAG, and a PaperQA2-style pipeline, ScholAR matches the strongest within bootstrap 95% CIs with a simpler design, and on unanswerable questions the local models abstain (18–20 of 20), each within an 18 GB budget. We also contribute a 100-case, 25-paper benchmark; a designed human evaluation is pending.

1 Introduction

Reading an unfamiliar paper is slowest at the moments that matter most: checking that a stated number, a method detail, or a figure says what you think it says. General-purpose chat over the PDF helps with fluency but brings a new failure mode: it summarizes confidently while inventing page references and attributing quotes to places they do not appear, because a generic chat pipeline puts no hard constraint between a citation and the text the model was conditioned on.

Purpose-built scientific RAG systems close part of this gap but assume resources most readers lack. OpenScholar (Asai et al. 2024), PaperQA2 (Skarlinski et al. 2024), and SciRAG (Ding et al. 2025) depend on cloud-hosted frontier models, and OpenScholar on a datastore of tens of millions of papers. We ask a narrower question: what does grounded, citable question answering over a scientific document look like under a hard local-only constraint, with a single sub-15-billion-parameter model and a retrieval design simple enough to audit end to end?

Two ideas do the real work in ScholAR. **Page-preserving chunking** segments a document so that every retrieved chunk lies entirely within one PDF page, making a citation resolve to an exact, highlightable location. **Indirect citation grounding** assigns each retrieved passage a short-lived evidence identifier and instructs the model to cite only those identifiers, never a free-form page number; the application then rewrites each identifier into a numbered reference that maps deterministically to a page. The model decides only which evidence supports a claim, and the evidence-to-page mapping is application logic rather than a model guess. Beyond single-document text, ScholAR routes figure and table questions to a locally run multimodal model and extends to a paper's own cited references for cross-document questions.

Our contributions are:

- A local-inference RAG system for scientific documents that runs both text and vision on-device (paper acquisition and reference resolution use public arXiv/Semantic Scholar lookups by identifier), using page-preserving chunking and an indirect citation mechanism that rules out *invented* page numbers by construction (though not, we find, whether the cited page supports the claim).

- An evaluation that scores the faithfulness of the system's *generated answers* rather than of pre-written gold claims, using a local entailment judge validated by a negative control against the cosine proxy it replaces. Under the judge, faithfulness is a modest 0.61 across four local models with contradictions in about a quarter of answers, well below what a similarity proxy reports.
- Automated benchmarks and results for retrieval, retrieval-support faithfulness, visual grounding, and cross-document localization, the last measured against the published baselines of M3SciQA on its own data, where a locally resolved figure lifts localization from near-chance to competitive with a frontier cloud model.
- A 100-case benchmark across 25 papers spanning text, mathematical, and figure questions, with gold answers verified against their source passages. A model-agnostic human-evaluation protocol over these cases is described in Section 7; its expert scoring is pending, so we do not list it among our results.

2 Related Work

Retrieval-augmented generation. RAG conditions generation on retrieved evidence (Lewis et al. 2020). Lexical BM25 (Robertson and Zaragoza 2009) remains a strong baseline that zero-shot dense retrievers often fail to beat (Thakur et al. 2021), which is why ScholAR keeps BM25 primary and treats dense retrieval as a fusion component.

Scientific-literature RAG systems. PaperQA and PaperQA2 (Lála et al. 2023; Skarlinski et al. 2024) are agentic systems with frontier backends; OpenScholar (Asai et al. 2024) synthesizes over a large open datastore, and SciRAG (Ding et al. 2025) adds outline-guided synthesis. All three assume cloud-hosted or large self-hosted models; ScholAR instead targets a single local sub-15B model, trading capability so that no document content leaves the machine (only public arXiv/S2 identifier lookups touch the network). Closest to our citation mechanism, Pleias-RAG (Pleias 2025) *trains* small models to emit source quotes; we instead put the grounding in the retrieval and application layer, so it works with any local model rather than one taught to quote.

Citation and faithfulness evaluation. Gao et al. (2023) formalize citation precision and recall via entailment; SummaC (Laban et al. 2022), FActScore (Min et al. 2023), and RAGAS (Es et al. 2024) score consistency, atomic-fact support, and RAG faithfulness respectively. Our generation-faithfulness metric uses FActScore-style atomic decomposition scored by a local LLM entailment judge on the system's own answers; a lighter cosine signal is used only for retrieval-support scoring.

Visual and multi-document scientific QA. ColPali (Faysse et al. 2024) indexes pages as visual embeddings without OCR; ScholAR keeps text retrieval primary and routes only figure and table queries to a cropped-region vision call. M3SciQA (Li et al. 2024) finds even large models trail human experts at cross-paper reasoning; we evaluate directly on it (Section 6.6). GROBID (Lopez 2009) and Nougat (Blecher et al. 2023) target structured extraction, complementary to our chunking.

3 Problem Formulation

Given a scientific PDF D, a question q, and extracted chunks $\mathcal{C} = \{c_1, \ldots, c_n\}$ (in multi-document mode also from D's cited references), the system retrieves a ranked subset $\mathcal{C}_q = \mathrm{Retrieve}(q, \mathcal{C}, k)$ and generates a grounded answer. Each chunk c_i carries a source page number and, across documents, a source paper identifier, so $\mathcal{C}_q$ resolves to a specific page of a specific document. Generation attaches each claim to an evidence identifier referencing a specific $c_i \in \mathcal{C}_q$ rather than a free-form page number, so citations are constrained to what was actually retrieved, independent of $\mathrm{Retrieve}$'s own precision.

4 Method

Figure 1 shows the end-to-end pipeline. A PDF is chunked one page at a time, ranked by BM25, and exposed to the model as evidence identifiers; the model's identifier citations are then rewritten into page-anchored references by the application rather than by the model.

4.1 Document Processing and Page-Preserving Chunking

Given an arXiv identifier or an uploaded PDF, ScholAR extracts per-page text with PyMuPDF (Artifex Software 2024) and persists metadata, pages, chunks, and figures as JSON. Standard character or token windows routinely split a chunk across a page boundary, breaking the one-to-one map between a retrieved chunk and the page a citation should highlight. ScholAR instead chunks strictly within each page: it slides a window of target size 1400 words with 120-word overlap over each page's tokens independently, so a chunk never spans two pages. A lightweight regular-expression pass tags each chunk with a detected section title when a heading is found near the top of the page. Chunk identifiers are numbered per paper.

4.2 BM25-Primary Retrieval

The production retriever ranks chunks with Okapi BM25 (Robertson and Zaragoza 2009) (k_1=1.4, b=0.72) as the primary signal, then applies small additive heuristic boosts: a query-expansion dictionary maps research terms to related vocabulary, an explicit page number in the query adds a page-hint boost, and query language implying a section or a claim-bearing phrase adds a small boost. A separate hybrid configuration fuses BM25 with dense retrieval over a local all-MiniLM-L6-v2 sentence encoder (Reimers and Gurevych 2019) using Reciprocal Rank Fusion (Cormack, Clarke, and Buettcher 2009) (k=60); this configuration is used in the faithfulness study.

4.3 Indirect Citation Grounding

Asking a model to write a page number directly (for example "[p. 7]") is fragile, because the model has no reliable way to know which page a claim came from and will produce a plausible-looking one. ScholAR instead assigns each retrieved passage a short-lived evidence identifier for the current turn

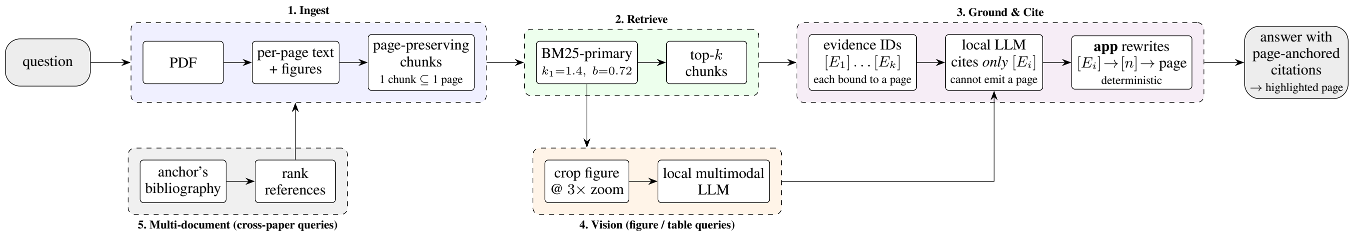

Figure 1: The ScholAR architecture. A question enters at the left and leaves as an answer whose every citation resolves to a highlightable page. **(1)** Ingest keeps each chunk inside a single PDF page. **(2)** BM25-primary retrieval returns the top-k chunks. **(3)** Those chunks are exposed to the model only as evidence identifiers $[E_i]$, each already bound to the page of the chunk it came from; the model may cite an identifier but can never write a page number, and the *application*, not the model, rewrites $[E_i]$ into a numbered reference and resolves it to a page. A cited page is therefore correct by construction rather than by the model's good behavior. **(4)** Figure and table queries additionally route a cropped, 3×-zoomed image to the same local multimodal model. **(5)** Cross-paper queries first rank the anchor's bibliography, and the selected reference is then ingested like any other paper. Every component runs on one machine.

and instructs the model to cite only these identifiers. After generation, the system discards any page number the model wrote on its own, rewrites each valid evidence identifier into a compact numbered reference in citation order, and resolves each reference to its source chunk's page for highlighting. The model decides only which evidence supports a claim; the mapping from evidence to page is deterministic application logic. This prevents page-number hallucination by construction rather than by asking the model to be careful.

4.4 Visual Grounding

When a query references visual content, through cue terms such as "figure" or an explicit figure number, retrieval boosts figure and table chunks so they can compete against longer text chunks. Captions are detected by regular expression and the region above each is rendered at 3× zoom and cached. A query naming an explicit figure number is matched against that figure's label directly, which matters because a letter-initial tokenizer collapses "Figure 2" and "Figure 14" to the same token once digits are dropped. When the top-ranked chunk is a figure or table, the cached image is sent with its caption to the same local model used for text (both of our multimodal models read it natively), never to a cloud API; if the image is missing, the system degrades to a caption-only answer.

4.5 Multi-Document Extension

ScholAR resolves an anchor paper's bibliography and lets the user ingest cited works into the same session. For arXiv papers, references are resolved by arXiv identifier; for uploaded PDFs, by a title search over the last pages, matched against numbered and author-year citation styles. Ingesting a reference applies the same page-preserving chunking and tags every chunk with a source paper identifier. At query time, chunks from all loaded papers merge into one retrieval pool, and each citation is labeled with its source paper.

4.6 Faithfulness Metrics

We use two measures for two questions, both over atoms decomposed in the FActScore style (Min et al. 2023). *Generation faithfulness* scores the system's actual answer against its retrieved context, classifying each atom entailment / neutral / contradiction with a local LLM entailment judge; we report the entailment fraction and a hallucination (contradiction) rate. We deliberately do not use embedding cosine here: cosine cannot separate a claim from its negation, and a negative control (Section 6.3) confirms the judge catches contradictions it misses. *Retrieval-support* instead scores a pre-written gold claim against the retrieved chunks (answerability) with a cheaper zero-shot cosine signal in the SummaC style (Laban et al. 2022); its Combined Faithfulness Score blends the cosine entailment fraction (0.50), whole-claim similarity SCR (0.30), and boundary-matched key-fact recall KFP (0.20), with the CFS≥0.55 cutoff calibrated on a small pilot. The generation judge is what our faithfulness claims rest on and the result we foreground.

4.7 System Architecture

The backend is a local application that stores parsed PDFs as JSON on disk; the frontend highlights cited pages. All inference, both text generation and figure or table question answering, runs locally; the evaluation swaps among four local models, two of which (qwen3.5:9b, gemma4:12b) are natively multimodal and handle figure and table queries. No cloud API is used for inference and no user data leaves the machine. Paper acquisition and reference resolution use public arXiv and Semantic Scholar lookups by identifier only. All experiments in this paper were run on a consumer laptop (Apple Silicon, 18 GB unified memory) under Python 3.12, with all-MiniLM-L6-v2 loaded from a local cache; no discrete or cloud GPU was used, and inference runs on the laptop's integrated GPU through the local model server's accelerated backend.

5 Benchmarks

We use two families of evaluation data. The first is a set of small, manually labeled benchmarks over three landmark NLP papers (Attention Is All You Need (Vaswani et al. 2017), Retrieval-Augmented Generation (Lewis et al. 2020), and

Table 1: Key statistics of the ScholAR corpus and benchmarks. Page-preserving chunking yields close to one chunk per page, because a typical page holds fewer words than the chunk window.

Corpus		*Diverse benchmark*	
Papers	25	Questions	100
Pages / paper	34.5	text/math/vis.	50/25/25
Chunks / paper	34.8	Question words	12.9
Words / paper	17,694	Gold facts / q.	2.8
Labeled sets (hand / auto)		*Generated answers*	
Retrieval	14 / 100	Answers	350
Faithfulness	51 / 100	Local models	4
Visual / multi-doc	18 / 18	Inline citations	829
Unanswerable	20	Citations / answer	2.4

LLaMA (Touvron et al. 2023)): a 14-case retrieval benchmark with labeled relevant chunks, a 51-case faithfulness benchmark with oracle claims and a supporting chunk, an 18-case visual-grounding pilot, and an 18-case multi-document benchmark of which 10 have an arXiv-resolvable target. These support the automated metrics that need ground-truth chunk labels, and they are small: the hand-labeled benchmarks cover three papers between them.

The second is a 100-case benchmark spanning 25 papers, built to broaden coverage and reduce the memorization advantage a model has on very famous papers. It contains 50 text, 25 mathematical, and 25 figure or table questions. Each case was mined from a real passage or caption by a local model and then kept only if the required answer facts appear, on a word or number boundary, in the source passage; questions from benchmark-dataset papers, reference lists, and worked examples were filtered out. Because each case carries exact substrings copied from its source passage, we recover a gold supporting chunk for every case at no extra labeling cost: it is the chunk that contains those substrings, disambiguated by the recorded page. This yields a 100-case, 25-paper labeled set for retrieval and retrieval-support faithfulness, seven times the 14-case retrieval set and eight times the papers of the hand-labeled set. The labels are weaker than hand annotation, and the questions, being mined from passages, carry lexical overlap that can favor lexical retrieval; we therefore report the scaled set alongside the 3-paper hand-labeled set rather than in place of it. This diverse benchmark is also the dataset for the human evaluation (Section 7). Table 1 summarizes the corpus and the benchmarks built on it.

6 Automated Evaluation

6.1 Retrieval

Table 2 reports retrieval on the 100-case scaled set. The system we deploy, BM25-primary (BM25 with lightweight heuristic reranking), reaches Recall@5 0.93 and MRR 0.861; plain BM25 is within noise (0.94, 0.863), and a keyword-overlap baseline even edges Recall@5 (0.95), so no lexical variant is clearly best. Table 3 isolates the reranking heuristics: layering semantic and section boosts onto plain BM25 costs 0.002 MRR, and the page-hint boost the system actually

Table 2: Retrieval on the scaled benchmark: 100 auto-labeled cases across 25 papers, with NDCG@5 over the gold chunks. Here and in later tables, **best** and second per column are marked. On the 3-paper hand-labeled anchor (14 cases) dense-only instead leads (R@5 1.000, MRR 0.881).

System	R@1	R@3	R@5	MRR	NDCG@5
Keyword	0.670	0.860	**0.950**	0.779	0.762
BM25-Only	**0.810**	**0.910**	0.940	**0.863**	**0.828**
BM25-primary	**0.810**	**0.910**	0.930	0.861	0.824
Dense-Only	0.470	0.650	0.740	0.572	0.523

Table 3: Retrieval ablation on the 100-case set: the marginal effect of layering each reranking heuristic onto plain BM25. Neither layer improves aggregate retrieval, and the page-hint boost the system deploys changes nothing measurable.

Configuration	R@5	MRR	NDCG@5
Plain BM25	0.940	0.863	0.828
+ semantic/section rerank	0.930	0.861	0.824
+ page-hint boost (deployed)	0.930	0.861	0.824

deploys adds exactly 0.000, so the heuristics earn their keep on robustness, not on the aggregate numbers. The clear loser is single-pass dense retrieval (0.74, 0.572): on 25 diverse papers it trails every lexical configuration, the expected BEIR outcome (Thakur et al. 2021) and the reverse of the 3-paper anchor where dense led. We read BM25's margin cautiously, since the mined queries share lexical surface with their source passages, but the direction, lexical over single-pass dense at scale, is the one BM25-primary assumes.

6.2 Retrieval-Support Faithfulness

Table 4 reports the retrieval-support CFS on the scaled 100-case set. BM25-primary and the hybrid BM25+Dense+RRF land within 0.003 (CFS 0.785 vs 0.782), and BM25-primary has the higher supporting-chunk hit rate (SCHR@5 0.86 vs 0.75); the hybrid's narrow edge on the 3-paper anchor (0.827 vs 0.807) does not survive at scale, since a single-pass dense encoder weak on diverse papers adds nothing, which is why BM25-primary is the default. This measures whether retrieval surfaces support for a known claim, not whether the generated answer is faithful (that is Section 6.3).

6.3 Generation Faithfulness

We decompose each generated answer into atomic claims and ask whether the retrieved evidence entails each one, using a local LLM entailment judge (qwen3.5:9b), not a sentence-embedding similarity: cosine cannot separate a claim from its negation and so structurally never reports a contradiction. A negative control confirms this: on gold claims deliberately corrupted (numbers perturbed, relations negated), a cosine proxy flagged 50% as unfaithful and fired a contradiction on 0%, whereas the judge flagged 90% and fired a contradiction on 75%. Scored with the judge over the 100 diverse cases and four models (Section 6.4), answers reach a

Table 4: Retrieval-support faithfulness on the scaled 100-case set (25 papers). Semantic is the cosine entailment fraction; SCR is whole-claim cosine; KFP is key-fact recall; SCHR@5 is the supporting-chunk hit rate at depth 5. This retrieval-support signal is cosine-based, unlike the generation judge of Section 6.3.

Metric	BM25-primary	Hybrid+RRF
Semantic (cosine)	0.925	0.915
SCR (cosine)	0.433	0.438
KFP (key-fact recall)	0.966	0.966
Combined CFS	0.785	0.782
SCHR@5	0.860	0.750
Faithful (≥ 0.55)	93/100	92/100

mean faithfulness of 0.61, far below the 0.85 macro-mean a cosine proxy reports on the same four models (0.94 on the 3-paper single-model anchor); about a quarter of answers (93 of 350) contain a contradicted atom, and of 1,279 citation occurrences checked (a source cited in several sentences is checked each time; Table 1 counts the 829 distinct markers), 73% are supported, 18% partial, and 9% unsupported, against the "none unsupported" cosine had claimed. These numbers are lower and honest; validating them against expert judgment is the human evaluation's job (Section 7).

6.4 Does the Grounding Hold Across Models?

We swap only the generation model and re-score the pipeline over the 100 diverse cases with the same entailment judge (Table 5). Under the judge the picture is both lower and flatter than a cosine proxy suggested. Generation faithfulness is uniformly modest across the four models (0.59 to 0.65): no local model is markedly more faithful, so grounding is model-agnostic in the weak sense that swapping the model changes little, though the level is mediocre. Citation support varies more (0.64 to 0.86), inversely with how freely a model cites: qwen3.5:9b emits the most citations (633) but supports the fewest (0.64), whereas gemma4:12b cites less and more precisely (354, 0.86). A cosine proxy reported almost the opposite (faithfulness spread 0.72 to 0.95, support a tight 0.87 to 0.95); the negative control of Section 6.3 is why we trust the judge's lower, flatter version.

6.5 Visual Grounding

On the 18-case visual pilot, retrieval routes to the correct figure or table in all 18 cases, never falling back to the caption-only path. We do not report our earlier keyword-overlap answer-quality proxy: it rewards lexical overlap rather than visual correctness, so it cannot tell whether the image was read. The cleaner evidence that vision matters is Section 6.6, where resolving the figure is worth 2.6× on MRR.

6.6 Multi-Document Localization

We evaluate cross-document localization on M3SciQA (Li et al. 2024), whose *locality* task is exactly ours: given an anchor paper and a question about one of its figures, rank the anchor's bibliography (a mean of 47.9 candidates) to find

Table 5: The same ScholAR pipeline with only the generation model changed, scored by the local entailment judge. The two multimodal models run all 100 diverse cases; the text-only models (llama3.1, mistral) run the 75 non-visual cases, so counts differ (350 answers, not 400). Faith. is the fraction of answer atoms entailed; Hall. the fraction judged *contradiction*; Cites the inline citations checked, Supp. the fraction supported. Faithfulness is uniformly modest across models; citation support varies.

Model	Faith.	Hall.	Cites	Supp.
gemma4:12b	**0.645**	<u>0.096</u>	354	**0.856**
qwen3.5:9b	0.594	**0.094**	633	0.643
mistral:7b	0.602	0.169	213	0.718
llama3.1:8b	<u>0.607</u>	0.140	79	<u>0.835</u>

the reference that answers it. Their 297 gold-labeled locality cases and metric place us against their published baselines (Table 6).

Text-only retrieval is barely above chance. BM25-primary reaches MRR 0.180 against an empirical random floor of 0.121 on the pools we rank, and dense MiniLM (0.125) is indistinguishable from random; M3SciQA report 0.127 for BM25 against a floor of 0.126, so our setup reproduces the shape of theirs. This also explains our own 18-case benchmark, where BM25 scored 0.183: the question names an entity that appears only inside a figure, so a signal over titles and abstracts has almost nothing to match against.

The bottleneck is therefore not retrieval but resolving what the question points at. Routing the anchor figure through the same local multimodal model first, then retrieving with the resolved entity, lifts MRR to 0.474 (qwen3.5:9b) and 0.455 (gemma4:12b): 2.6 times the text-only pipeline and 3.3 times the strongest open-source multimodal baseline they report (0.144), ahead of GPT-4V (0.400), and within 0.026 of GPT-4o (0.500), on a laptop. Expert humans remain far ahead at 0.796, so the task is not solved. Two caveats keep this honest. The daggered rows are M3SciQA's figures on their 1000-item test split, whereas we use their 297 labeled locality cases and rank a pool our resolver reconstructs (47.9 vs their 42.8 references), so our random floor (0.121) differs from theirs (0.126): a shared task and metric, not an identical set. And our pipeline decomposes the task (vision then BM25); but so do M3SciQA's simple baselines (GPT-4o caption, then embed and rank), so text-embedding-3-large (0.297) already uses a frontier cloud model for vision, and our 0.474 beats it with a local one.

6.7 Comparison with Local Baselines

To place ScholAR against alternatives under its own local constraint, we reimplement three baselines on the same model (qwen3.5:9b) and the same cases (51 labeled and 40 diverse): long-context PDF-chat, which puts the whole paper in the model's context; vanilla RAG, with fixed cross-page chunks and dense retrieval; and a local reimplementation of the rerank-and-summarize step of PaperQA2 (Skarlinski et al. 2024). In Table 7 ScholAR tops generation

Table 6: Multi-document localization on M3SciQA's locality task (297 labeled cases, mean 47.9 candidate references). Rows marked † are figures published by M3SciQA (Li et al. 2024); the rest are measured here. **Best** and <u>second-best</u> *system* per column are marked; human and random are reference rows, not ranked. Resolving the figure with a local multimodal model, rather than retrieving from the question alone, is what moves this task.

System	MRR	R@5
Human expert†	0.796	–
GPT-4o† (cloud)	**0.500**	–
ScholAR + vision (qwen3.5)	<u>0.474</u>	**0.606**
ScholAR + vision (gemma4)	0.455	<u>0.572</u>
GPT-4V† (cloud)	0.400	–
text-emb-3-large† (cloud)	0.297	–
ScholAR BM25 (text only)	0.180	0.242
Best open-source LMM†	0.144	–
BM25† / Random†	0.127 / 0.126	–

Table 7: ScholAR versus local baselines on the shared model (qwen3.5:9b) and cases (51 labeled + 40 diverse). Faith. is generation faithfulness; Correct. is must-include recall; Cite-F1 is citation F1 in the sense of Gao et al. (2023). All systems emit in-range page numbers (invalid-page rate 0); this checks only that a cited page exists, not that it supports the claim, which we examine separately below.

System	Faith.	Correct.	Cite-F1
PDF-chat (long-context)	0.801	0.267	0.742
Vanilla RAG	0.860	0.340	**0.802**
PaperQA2-style (RCS, local)	<u>0.872</u>	<u>0.550</u>	<u>0.782</u>
ScholAR	**0.892**	**0.563**	0.760

faithfulness (0.892) and answer correctness (0.563), but the margin over the PaperQA2-style system is not resolvable on 91 cases (paired 95% CIs $+0.020\,[-0.020, +0.059]$ and $+0.013\,[-0.104, +0.133]$, both spanning zero). ScholAR therefore *matches* a PaperQA2-style pipeline with a simpler design, while both clearly beat long-context PDF-chat (correctness 0.267): retrieval beats putting the whole document in context. We compare against local reimplementations, not the hosted OpenScholar, PaperQA2, or SciRAG; a resource-matched local comparison, not a claim to frontier-scale quality.

Do cited pages support their claims? The invalid-page rate checks only that a cited page exists. Judging with the entailment judge whether each cited page actually supports its claim, ScholAR's pages do so 66% of the time and long-context PDF-chat's 71%: the motivating failure is real (about a third of cited pages, for both systems, do not support their claim), and indirect grounding does not fix it. The mechanism guarantees a cited page is real and was retrieved (invalid-page rate 0), not that the page supports the claim, which depends on retrieval, where ScholAR is no better than PDF-chat. We therefore read it as a provenance and auditability guarantee, not a faithfulness improvement.

Table 8: Per-model behavior of the unmodified ScholAR pipeline: only the generation model changes. Abstain / Fabricate are measured on the 20 provably-unanswerable questions (Abstain is the fraction correctly declined). Mem. is the loaded footprint in GB (weights plus KV cache) and latency is end-to-end in seconds (mean / p95) over 20 questions on the evaluation laptop; retrieval adds about 40 ms for every model.

	Unanswerable		**Cost per query**		
Model	**Abst.**	**Fabr.**	**Mem.**	**Tok/s**	**Lat.**
qwen3.5:9b	1.00	0.00	6.1	21.9	11.4/14.6
gemma4:12b	0.95	0.05	8.1	16.3	8.8/14.2
llama3.1:8b	1.00	0.00	6.9	26.9	6.3/17.8
mistral:7b	0.90	0.10	6.5	24.9	6.1/8.4

6.8 Abstention on Unanswerable Questions

A grounded system should decline when the answer is not present. We built 20 provably-unanswerable questions, each posed against a different paper in which the queried fact is provably absent (exact substring), so the correct response is to abstain. All four models decline on the large majority (Table 8): qwen3.5:9b and llama3.1:8b on every question, gemma4:12b on 19 of 20, mistral:7b on 18. The few non-abstentions are a benign overlap, not invention (a wrongly paired paper that independently reports a minibatch size). The set is small and synthetic (n=20, so all four models are statistically indistinguishable) and we ran no ungrounded control, so we read this as a tendency of the grounded pipeline, not a causal claim or a precise rate.

6.9 Local Deployability

We also timed the answering path over 20 questions per model on the same laptop (Apple Silicon, 18 GB). Retrieval is not the bottleneck: BM25 returns in about 40 ms regardless of model, so latency is generation-bound. Every model fits the 18 GB budget (6 to 8 GB loaded) and answers in single-digit to low-double-digit seconds (Table 8). No cloud call and no GPU cluster: one laptop runs the loop.

7 Human Evaluation (In Progress)

Automated metrics measure component behavior, not whether a reader would judge an answer correct and well-cited. Following the practice of comparable systems (Asai et al. 2024; Ding et al. 2025; Skarlinski et al. 2024), we designed a model-agnostic human evaluation over the 100-case diverse benchmark (Section 5). Each question is answered by the four local models (qwen3.5:9b, gemma4:12b, llama3.1:8b (Llama Team 2024), mistral:7b (Jiang et al. 2023)) running the *same* pipeline, so only the generation model changes. Blind and in randomized order, expert evaluators score every answer on four anchored 1–5 dimensions (Relevance, Coverage, Faithfulness, Usefulness) and grade each inline citation Supported

Table 9: Human-evaluation results (in progress). Values are placeholders pending expert scoring; they are not estimated. Dimensions are mean 1–5.

Model	Rel.	Cov.	Faith.	Use.	Cite-F1
qwen3.5:9b	–	–	–	–	–
gemma4:12b	–	–	–	–	–
llama3.1:8b	–	–	–	–	–
mistral:7b	–	–	–	–	–

/ Partial / Unsupported. From these we compute per-model means, citation precision, recall and F1, a Friedman test of whether grounding is model-agnostic, and the correlation between human faithfulness and our automated metric. All 350 answers are generated; expert scoring is the remaining step, so Table 9 reports placeholders, not estimates.

Once available, a Friedman test across models and the human-to-automated Faithfulness correlation (both to be filled) will test whether grounding holds across local models and whether our judge tracks expert judgment.

8 Discussion and Limitations

ScholAR's contribution is a local-inference integration whose two elements, page-preserving chunking and indirect citation grounding, are simple, auditable, and make a cited page correct by construction. Under a real entailment judge, though, the generated answers are only modestly grounded (faithfulness 0.61, contradictions in about a quarter of answers), and the rest of this section is where our evidence runs thin. The retrieval and retrieval-support benchmarks span 100 cases across 25 papers, but their gold chunks are auto-derived from mined facts whose lexical overlap likely flatters lexical retrieval; the 3-paper hand-labeled sets remain the higher-precision anchor. The visual (18 cases) and abstention (20 synthetic negatives) probes are small, so their deltas carry no significance testing and indicate tendencies, not guarantees. And the generation-faithfulness judge, though it catches contradictions a cosine proxy misses (our negative control), is itself calibrated on a small pilot and unvalidated against human ratings; closing that gap is what the human evaluation exists for.

Our comparison (Section 6.7) is against local reimplementations of PDF-chat, vanilla RAG, and PaperQA2's rerank-and-summarize step, not against the hosted OpenScholar, PaperQA2, or SciRAG. Those systems depend on cloud-hosted models and large datastores, so a head-to-head would require either a cloud run, contradicting our local-only design, or crippling them to an untested backend; the honest scope of our result is therefore resource-matched local systems, not frontier-scale quality. Finally, cross-document localization is competitive only once the anchor figure is resolved by the vision model; from the question text alone it sits near chance, and even our best configuration (MRR 0.474) remains far below the 0.796 of expert humans on the same task, so the cross-document setting stays the weakest part of the system.

9 Conclusion

We presented ScholAR, a local-inference retrieval-augmented system for grounded scientific document comprehension combining page-preserving chunking with indirect citation grounding, extended to visual and multi-document questions. Under a validated local entailment judge its generated answers reach a modest faithfulness of 0.61 across four models (contradictions in about a quarter), well below what a cosine proxy reports; retrieval reaches Recall@5 0.93 across 25 papers; and it matches a PaperQA2-style pipeline on correctness and faithfulness within CIs, with a simpler design. Indirect citation grounding guarantees a cited page is real and was retrieved, but a page-support check (66%, no better than a free-form baseline) shows it does not make the page support the claim: a provenance guarantee, not a faithfulness fix. On M3SciQA, resolving the anchor figure locally lifts localization to MRR 0.474, ahead of every open-source baseline there and close to GPT-4o but far below expert humans. We also contribute a 100-case, 25-paper benchmark.

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